Unsupervised Deep Learning for Ecological Site Groups in MLRA 42B Progress Report (PS5)

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Progress since PS4

The full pipeline now runs end to end on a GPU in about 6 minutes: data preparation, autoencoder training, clustering for k = 3–12, a center-pixel baseline, and a point export. Training converged without overfitting (validation MSE 0.14 in z-score units). However, we found that roughly 28% of MLRA 42B appears to be excluded by a data gap, and diagnostics suggest the autoencoder may mostly encode patch averages rather than spatial pattern. Fixing these comes before choosing k or comparing with mapped ecological sites.

Problem statement

Across the western United States, land managers, ranchers, and conservation agencies rely on a system of land classification called Ecological Sites. An Ecological Site is a patch of landscape with a distinct combination of soil, climate, and topography that determines what plants can grow there and how the land responds to disturbance. It describes the whole soil-climate-vegetation system and how it can shift between different states over time (Karl and Herrick, 2010; Bestelmeyer et al., 2003). Ecological site maps are built largely by hand: soil scientists and ecologists visit sites, take measurements, and use expert judgment to draw boundaries and assign classifications. This is accurate but slow and expensive, and in practice ecological site maps are produced by linking ecological sites to soil survey (SSURGO) map units rather than being mapped independently.

The ecological site concept has become cumbersome; many sites with different names have properties that are functionally quite similar. The USDA would like to transition from ecological sites to ecological site groups, effectively absorbing ecological sites with similar properties into broader classification categories. This project asks whether an unsupervised deep learning model can divide MLRA 42B (Southern Rio Grande Rift) into ecologically meaningful groups using only environmental covariates. We then ask where those groups agree and disagree with mapped ecological sites. Disagreement is the main interest: it may give insight into existing map-unit uncertainty, ecological site concepts that don't fit the landscape, or gaps in our covariates or model.

Motivation and objectives

Ecologically important patterns are spatial (landform position, drainage, soil properties) and are lost when pixels are treated independently (Padarian et al., 2019; Wadoux et al., 2019). The goal is to discover groups without labels, which suits representation learning. Prior digital ecological-site mapping has been supervised and trained on existing ecological classes (Maynard et al., 2019). Our unsupervised approach does not assume current ecological site concepts capture all the heterogeneity of landscape patterns. The task type is unsupervised clustering.

Dataset description

The final stack has 19 layers stored as 20 channels: aspect is now split into northness and eastness. The study extent is MLRA 42B plus a 5 km buffer. All data are public except the STEDIS terrain and spectral derivatives, which are derived from public data (USGS 3DEP elevation and Landsat imagery) but are not yet published. MLRA 42B is a region of desert basins, alluvial fan piedmonts, and isolated mountain ranges in southern New Mexico and west Texas, where carbonate accumulation, soil depth, texture, and water redistribution strongly shape plant communities (Gile et al., 1981).

Table 1. Covariate sources and layers retained.

Group	Source	Native resolution	Candidate layers	Layers in stack	Channels
Terrain	STEDIS terrain derivatives (from USGS 3DEP DEM)	30 m	8	7	8
Spectral	STEDIS spectral derivatives (from Landsat)	30 m	3	3	3
Climate	PRISM 1991–2020 normals	~800 m	5	3	3
Soil	SOLUS (Nauman et al., 2024)	100 m	7	6	6
Total			23	19	20

Table 2. The 20 channels. Skewed layers were log10-transformed before z-scoring. "_32" layers use a 32-pixel (~960 m) neighborhood; "_10" and "_5" are wetness and channel-distance parameters; SOLUS names give property, depth (cm), and estimate type.

Channel	Group	What it is	Transform	Why it is used
sl_32	Terrain	Slope	log10	Runoff vs. infiltration, erosion, soil depth; a standard ecological site criterion
aspct_32 northness	Terrain	North–south component of aspect	Circular (from degrees)	Solar exposure, soil temperature, and moisture on mountain and hill slopes
aspct_32 eastness	Terrain	East–west component of aspect	Circular (from degrees)	As above
perctelev_32	Terrain	Percentile elevation within neighborhood (0–100)	none	Relative landscape position, independent of absolute elevation

Channel	Group	What it is	Transform	Why it is used
relmeanelev_32	Terrain	Elevation minus neighborhood mean	none	Separates run-off (convex) from run-on (concave) settings
tpi_32	Terrain	Topographic position index (Weiss, 2001)	none	Ridge, slope, and valley position; r = 0.73 with relmeanelev_32
swi_10	Terrain	SAGA wetness index (Böhner et al., 2002; Conrad et al., 2015)	none	Water redistribution; draws, swales, playa margins
vdcn_5	Terrain	Vertical distance to channel network (Conrad et al., 2015)	none	Floodplains and low terraces vs. fan remnants and uplands
B5 (SWIR1)	Spectral	Band 5 of a 6-band Landsat change-free composite (~1.55–1.75 µm)	none	Soil surface brightness, mineralogy, gravel and carbonate surfaces
NDVI_90_IQR	Spectral	Interquartile range of NDVI over the time series	log10	Rain-responsive grassland vs. stable shrubland
NDVI_Trend_post 2000	Spectral	Linear NDVI trend since 2000	none	Long-term greening or browning; reflects current state, not potential
prism_ppt	Climate	Mean annual precipitation	log10	Primary limit on production; precipitation zones
prism_tmax	Climate	Mean daily maximum temperature	none	Evaporative demand; hot basins vs. cooler uplands
prism_solclear	Climate	Clear-sky solar radiation	none	Energy for evaporation; varies < 1% in 42B
anylithicdpt	Soil	Depth to lithic contact (cm)	none	Rooting depth and water storage; mean 191 cm, SD 13 cm, so mainly flags shallow soils
caco3_30	Soil	$CaCO_3$ equivalent at 30 cm	log10	Shallow calcic horizons

Channel	Group	What it is	Transform	Why it is used
caco3_150	Soil	$CaCO_3$ equivalent at 150 cm	none	Deep carbonate and petrocalcic horizons on older surfaces
claytotal_30	Soil	Clay at 30 cm	none	Water-holding capacity, argillic horizons; sandy vs. loamy vs. clayey sites
fragvol_0	Soil	Rock fragments at surface	none	Gravelly piedmonts and desert pavements
fragvol_30_h	Soil	Rock fragments at 30 cm, upper estimate	none	Subsurface water storage and rooting

Validation data (not model inputs). Ecological sites from SSURGO map unit components linked to Ecological Site Descriptions (EDIT).

SOLUS: SOLUS layers are model predictions partly trained on soil survey component estimates (Nauman et al., 2024), so they are not fully independent of the SSURGO-linked validation map. We plan to run the model with and without SOLUS to measure how much they drive agreement.

Spectral layers: The three spectral layers describe current condition (state), not site potential and the model will be tested with and without them to test their influence on clustering.

Preprocessing (completed)

See report_ps4.pdf in repository.

Model implementation (completed)

The preliminary model is a convolutional autoencoder followed by k-means, implemented in one Python script (03_autoencoder_first_test.py, PyTorch). An autoencoder compresses each patch into a short numeric summary (a 16-number "fingerprint") and is trained to rebuild the patch from it; fingerprints are then clustered into candidate ecological site groups.

Patches and spatial split:

- 10,000 patches of 32 × 32 pixels (~960 m), all 20 channels, center pixel inside the AOI, no missing cells.
- Each patch lies entirely within one 128 × 128-pixel (~3.8 km) block, so no patch straddles training and validation areas. 20% of blocks containing patches were held out (424 of 2,122 blocks; 2,016 validation patches).

Architecture (273,732 parameters):

- Encoder: three blocks of 3 × 3 convolution, batch normalization, ReLU, and 2 × 2 max pooling (32, 64, 128 filters), from 32 × 32 to 4 × 4, then a dense layer to 16 numbers.

- Decoder: a dense expansion, three blocks of nearest-neighbor upsampling plus convolution, and a final 3 × 3 convolution with linear output to 20 channels.
- No flip or rotation augmentation, because aspect channels change meaning when a patch is rotated.

Training: Mean squared error loss, Adam (learning rate 0.001; Kingma and Ba, 2015), batch size 128, learning rate halved after 3 epochs without validation improvement, early stopping after 8, and checkpoints every epoch so interrupted runs can resume.

Clustering and baseline:

- Fingerprints were z-scored, then clustered with k-means (10 random starts) for k = 3–12.
- For each k we record inertia (elbow), silhouette on a 5,000-patch sample (Rousseeuw, 1987), group sizes, and bootstrap stability: k-means refit on 20 resamples, each compared with the full fit by the adjusted Rand index (ARI; Hubert and Arabie, 1985).
- Baseline: PCA on center-pixel values (12 components, 91.4% of variance), then k-means with the same k, compared with the autoencoder groups by ARI.

Outputs: Loss curves, per-channel validation R^2, example reconstructions, clustering metrics and plots, fingerprints, a GeoTIFF of where complete patches can be centered, and a GeoPackage of patch centers with group labels for every k.

Compute environment: A department GPU server (Simurgh), accessed over SSH. The script installs its own dependencies on first run, and a CPU mode supports runs on a 16 GB PC.

Preliminary results

The first full run took 6.2 minutes (5.0 minutes of data preparation, about 43 seconds of training). Results below come from 10,000 patches and should be read as diagnostic, given the data gap described under Problems identified.

Training: Early stopping ended training at epoch 69, at 0.6 seconds per epoch. Validation loss fell from 0.42 to 0.14 MSE (z-score units) and tracked training loss throughout, so the model is not overfitting. An MSE of 0.14 on z-scored data corresponds to about 86% of variance reconstructed overall.

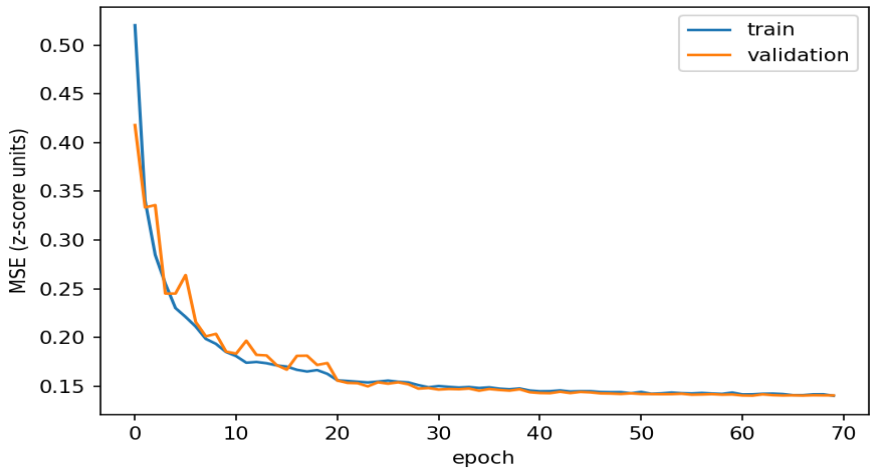


Figure 1. Training and validation loss (MSE, z-score units) by epoch. The two curves track each other, so the model is does not seem to be overfitting.

Per-channel reconstruction. Validation R^2 ranged from 0.22 (TPI) to 0.95 (maximum temperature). Climate, carbonate, texture, and slope were captured well; soil depth (0.39) and TPI (0.22) were mostly lost. Which means cluster probably say little about shallow soils or local landform position.

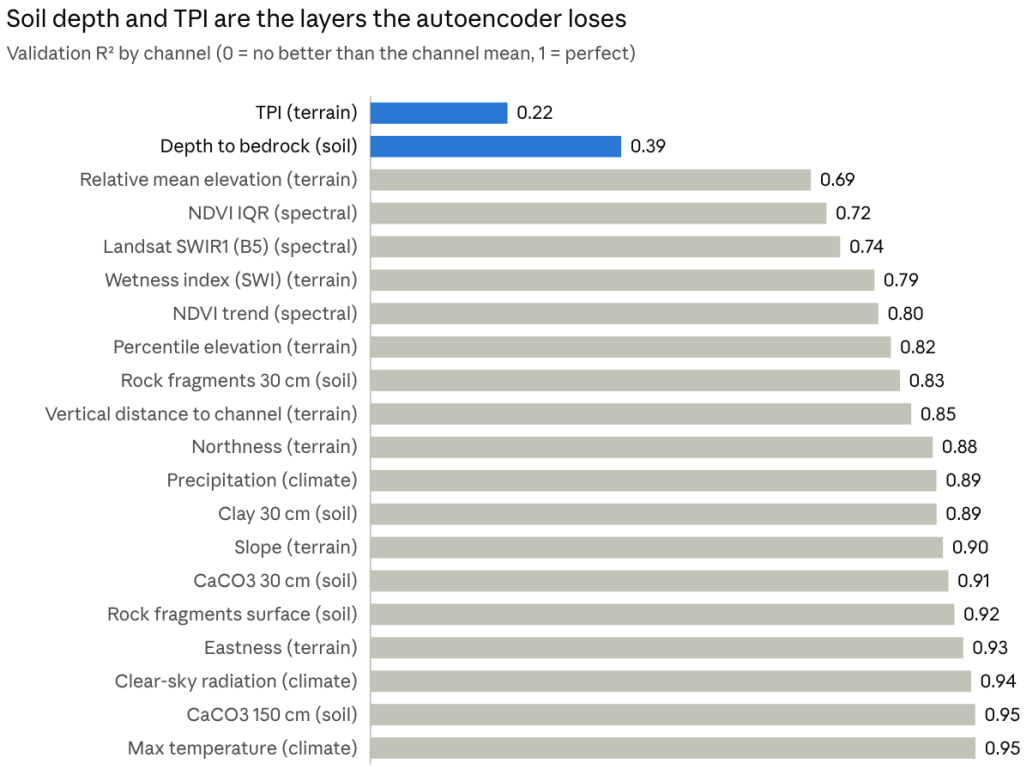


Figure 2. Validation R^2 for each of the 20 channels, lowest first. Bars below 0.5 are highlighted.

Reconstruction examples. Rebuilt patches show frame artifacts at patch edges, likely from zero padding, and lose fine texture (for example, NDVI IQR). Rebuilt slope patches show blotches absent from the smooth originals. Pull size resconstruction_samples.png can be viewed in output Test 1 folder on repository.

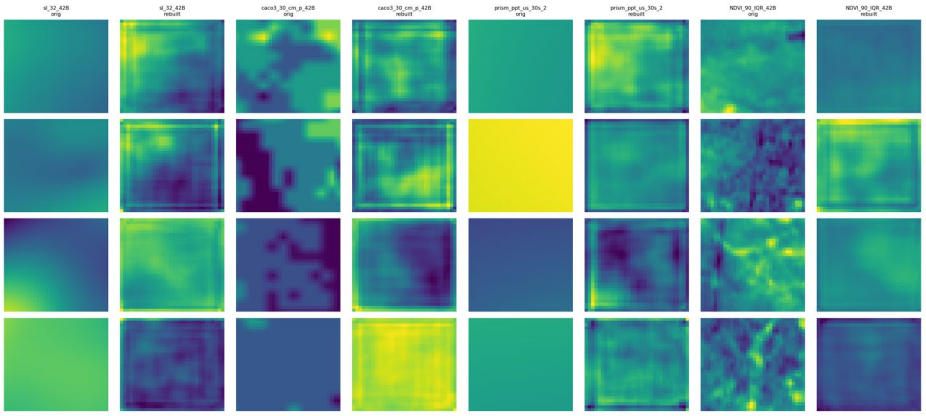


Figure 3. Four validation patches (rows): original and rebuilt for slope, $CaCO_3$ at 30 cm, precipitation, and NDVI IQR. Colors are stretched per patch, which exaggerates errors in near-uniform patches such as precipitation.

Clustering:

- Silhouette stayed between 0.156 and 0.170 for every k, below the 0.25 rule of thumb for substantial structure (Kaufman and Rousseeuw, 1990), and the elbow curve has no clear bend. The landscape appears to vary continuously.
- Bootstrap stability (mean ARI) was highest at k = 10 (0.89 ± 0.07).
- From k = 7 upward, one small group of 72–78 patches (under 1%) persist.
- Agreement between autoencoder and baseline groups was moderate (ARI 0.21–0.49), rising with k. The two methods group the landscape differently, but this does not show which is more ecologically meaningful. Silhouettes of the two methods are not comparable, because they are measured in different spaces.

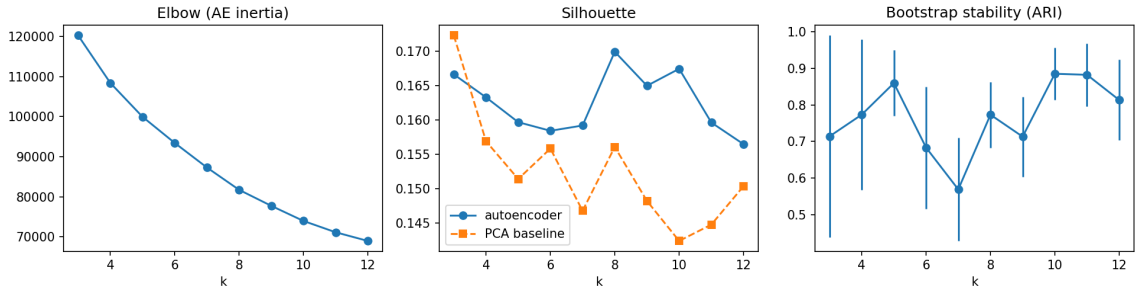


Figure 4. k-means diagnostics for k = 3–12: inertia (left), silhouette for the autoencoder and PCA baseline (center), and bootstrap stability as mean ARI ± 1 SD (right).

Problems identified

#	Problem	Evidence	Why it matters
1	About 28% of MLRA 42B may be excluded	$CaCO_3$ at 150 cm missing in 28.4% of the AOI; mask shows 100% complete; 27%	If the gap matches particular landforms (e.g., old surfaces with petrocalcic

#	Problem	Evidence	Why it matters
		of patch positions incomplete	horizons, shallow soils), the groups are biased toward the rest of the landscape
2	The autoencoder may mainly encode patch averages	Reconstructions lose within-patch texture; R^2 includes between-patch variation; several _32 layers are already smoothed over a patch-sized window	If a simpler non-spatial method would give similar groups, weakening the case for a CNN
3	Edge artifacts in reconstructions	Bright or dark frames on all rebuilt patches	Maybe a zero padding issue?
4	Soil depth and TPI poorly captured	Validation R^2 of 0.39 and 0.22	Shallow-soil and landform-position sites are ecologically very important but underrepresented. The depth problem may partly come from problem 1.
5	A persistent small group from k = 7 upward	72–78 patches	Could be a real rare landform (e.g., playas, rock outcrop) or an artifact. Will need to inspect in GIS.

Next steps

First, the data must be fixed for Caco3_150cm. We will change one thing at a time and compare per-channel R^2, stability, and ARI with the previous run.

1. **Diagnose the data gap:** Overlay the mask and the complete-patch map on the MLRA 42B boundary; compare the mask's AOI area (~32,900 km²) with the polygon area; check SOLUS documentation for how depths below restrictive layers are reported; overlay missing $CaCO_3$ at 150 cm on soil survey or geomorphic maps.
2. **Rebuild the stack:** Fill $CaCO_3$ at 150 cm according to step 1 (0%, a high value, or a separate "restrictive layer present" channel), or replace it with a restrictive-depth layer. Build the mask from the MLRA polygon only, never from data completeness.
3. **Improve the model and diagnostics:** Switch to reflect padding; add a within-patch R^2 (reconstruction of deviations from each patch's mean); add a second baseline of patch-mean covariates plus PCA and k-means; make the missing-data check cover the whole extent, not just the mask.
4. **Fix the rule for choosing k before rerunning:** Candidate rule: keep k with mean stability ARI ≥ 0.8 and smallest group ≥ 2% of patches, then pick the highest silhouette. Extend k to 20 as planned.
5. **Address poorly captured layers if needed:** consider loss weights or a larger fingerprint.
6. **Produce a wall-to-wall map:** by sliding-window prediction on the GPU.
7. **Describe each group:** by its covariate ranges, back-transforming log10 layers.

8. **Compare with SSURGO-linked ecological sites:** with and without SOLUS layers, and characterize where and why they disagree.

Code and reproducibility

All code is in the repository's src folder and runs in order: 01_clip_covariates.R (clipping), 02_build_covariate_stack.R (stack building), and 03_autoencoder_first_test.py (model, clustering, and export). README.md gives full commands, options, and outputs.

The data folder must hold the four outputs of the R scripts (stack, mask, channel order, z-score statistics); results are written to its outputs/ae_first_test subfolder.

- The Python script installs missing packages on first run and records every setting used (run_config.json) and a full log (run_log.txt). Any setting can be overridden from the command line.
- A fixed random seed (42) makes patch sampling and the train/validation split repeatable. GPU training is not bit-for-bit deterministic, so retrained models can differ slightly.
- The raster data are too large for the repository; the README lists the sources (PRISM, SOLUS, STEDIS, NRCS MLRA boundaries) and how to rebuild the stack.

References

Bestelmeyer, B. T., Brown, J. R., Havstad, K. M., Alexander, R., Chavez, G., & Herrick, J. E. (2003). Development and use of state-and-transition models for rangelands. Journal of Range Management, 56(2), 114–126.

Böhner, J., Köthe, R., Conrad, O., Gross, J., Ringeler, A., & Selige, T. (2002). Soil regionalisation by means of terrain analysis and process parameterisation. In E. Micheli, F. Nachtergaele, & L. Montanarella (Eds.), Soil Classification 2001 (European Soil Bureau Research Report No. 7, pp. 213–222). Office for Official Publications of the European Communities.

Conrad, O., Bechtel, B., Bock, M., Dietrich, H., Fischer, E., Gerlitz, L., Wehberg, J., Wichmann, V., & Böhner, J. (2015). System for Automated Geoscientific Analyses (SAGA) v. 2.1.4. Geoscientific Model Development, 8, 1991–2007. https://doi.org/10.5194/gmd-8-1991-2015

Gile, L. H., Hawley, J. W., & Grossman, R. B. (1981). Soils and geomorphology in the Basin and Range area of southern New Mexico: Guidebook to the Desert Project. New Mexico Bureau of Mines and Mineral Resources Memoir 39.

Hubert, L., & Arabie, P. (1985). Comparing partitions. Journal of Classification, 2, 193–218.

Karl, J. W., & Herrick, J. E. (2010). Monitoring and assessment based on ecological sites. Rangelands, 32(6), 60–64. https://doi.org/10.2111/RANGELANDS-D-10-00082.1

Kaufman, L., & Rousseeuw, P. J. (1990). Finding Groups in Data: An Introduction to Cluster Analysis. Wiley.

Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. International Conference on Learning Representations (ICLR). arXiv:1412.6980.

Maynard, J. J., Nauman, T. W., Salley, S. W., Bestelmeyer, B. T., Duniway, M. C., Talbot, C. J., & Brown, J. R. (2019). Digital mapping of ecological land units using a nationally scalable modeling framework. Soil Science Society of America Journal, 83(3), 666–686. https://doi.org/10.2136/sssaj2018.09.0346

McBratney, A. B., & de Gruijter, J. J. (1992). A continuum approach to soil classification by modified fuzzy k-means with extragrades. Journal of Soil Science, 43, 159–175.

Nauman, T. W., Brungard, C. W., Salley, S. W., & Allred, B. W. (2026). Image-based deep learning enables flexible, holistic 3D mapping of multiple soil properties. Geoderma, 473, 117985. https://doi.org/10.1016/j.geoderma.2026.117985

Nauman, T. W., Kienast-Brown, S., Roecker, S. M., Brungard, C., White, D., Philippe, J., & Thompson, J. A. (2024). Soil landscapes of the United States (SOLUS): Developing predictive soil property maps of the conterminous United States using hybrid training sets. Soil Science Society of America Journal. https://doi.org/10.1002/saj2.20769

Padarian, J., Minasny, B., & McBratney, A. B. (2019). Using deep learning for digital soil mapping. SOIL, 5, 79–89. https://doi.org/10.5194/soil-5-79-2019

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53–65.

Wadoux, A. M. J.-C., Padarian, J., & Minasny, B. (2019). Multi-source data integration for soil mapping using deep learning. SOIL, 5(1), 107–119. https://doi.org/10.5194/soil-5-107-2019

Weiss, A. (2001). Topographic position and landforms analysis [Poster]. ESRI User Conference, San Diego, CA.